Capstone: Autonomous Agent
Conceptual

Autonomous ICU Patient Monitoring Agent

Group 8

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Project Proposal and Scope

Use Case Review:

Intensive Care Units (ICUs) are high-stakes environments where patient conditions can deteriorate rapidly and unpredictably. Clinical staff must continuously monitor vital signs, heart rate, blood pressure, oxygen saturation (SpO2), respiratory rate, and temperature, across multiple patients simultaneously. This creates an enormous cognitive burden and increases the risk of delayed intervention.

Our proposed system is an Autonomous ICU Patient Monitoring Agent (AIPMA), an AI-driven autonomous agent deployed at the edge of the hospital network. The agent continuously processes multi-sensor vital sign streams, uses a generative AI model to detect early warning patterns and predict deterioration events before they become critical, and triggers real-time alerts to clinical staff with actionable decision support.

Generative AI Approach

The AIPMA leverages a hybrid generative and discriminative AI strategy:

• LLM Decision Support: A fine-tuned Large Language Model (LLM), specifically a medically-adapted transformer, acts as the decision-support engine, interpreting contextual alert summaries and producing clinician-readable explanations.

• GAN Data Augmentation: A Generative Adversarial Network (GAN) trained on synthetic ICU time-series data is used to augment training data for rare deterioration events (e.g., septic shock, arrhythmia onset), addressing class imbalance.

• Edge Autoencoder: A lightweight anomaly detection model (autoencoder) runs directly on the edge node for immediate, low-latency flagging of abnormal vital sign sequences.

Desired Outcomes

• Reduce mean time-to-alert for critical deterioration events by an estimated 40–60% compared to threshold-only monitoring systems.

• Decrease false positive alert rates using contextual AI reasoning, reducing alarm fatigue in nursing staff.

• Enable continuous 24/7 autonomous monitoring without requiring constant manual surveillance.

• Provide explainable AI outputs so clinicians understand why an alert was generated.

• Maintain full HIPAA compliance through on-premise edge processing so patient data never leaves the hospital network.

System Architecture

Architecture Overview

The AIPMA architecture follows a four-tier edge-to-cloud model: medical devices and bedside sensors form the perception layer; edge nodes at the ward level perform local inference; a hospital-internal aggregation server handles model orchestration; and a secure cloud layer (optional) supports model updates and longitudinal analytics.

Tier	Components	Function	Protocol
Tier 1: Sensors	Bedside monitors, ECG, SpO2, BP cuffs, temp probes	Raw vital sign data acquisition	HL7 FHIR / BLE
Tier 2: Edge Node	NVIDIA Jetson / Raspberry Pi 5 cluster per ward	Real-time inference, anomaly detection	MQTT (TLS 1.3)
Tier 3: Hospital Server	On-premise aggregation server, EHR integration	Model orchestration, alert routing, logging	HTTPS / REST API
Tier 4: Cloud (Optional)	Federated learning node, longitudinal analytics	Model retraining, population analytics	HTTPS / gRPC

Data Flow

The following describes the end-to-end data flow from sensor capture to clinician notification:

• **Step 1 – Sensor Capture:** Bedside medical devices sample vital signs at 1–250 Hz depending on sensor type and transmit structured readings via HL7 FHIR-compliant messages over the hospital's private network.

• **Step 2 – Edge Ingestion & Preprocessing:** The ward-level edge node ingests all streams via an MQTT broker. Data is preprocessed (normalization, noise filtering, missing-value imputation) by a lightweight Python microservice.

• **Step 3 – Anomaly Detection (Edge):** The autoencoder model analyzes a rolling 60-second window of multivariate vitals in real time. Reconstruction error above a dynamic threshold flags a potential anomaly.

• **Step 4 – Contextual AI Inference:** If an anomaly is flagged, a richer contextual inference is triggered: the LLM receives the patient's current vitals, recent trend summary, and relevant EHR context (age, diagnoses, medications) and generates a structured alert with severity level and clinical rationale.

• **Step 5 – Alert Delivery:** Alerts are delivered to the nurse station dashboard, mobile devices of assigned clinicians, and the hospital's EHR system via a REST webhook. All events are logged to an immutable audit trail.

Communication Protocols & Security Measure

• MQTT over TLS 1.3 between sensors and edge nodes: lightweight, low-latency, and encrypted.

• HTTPS with mutual TLS (mTLS) authentication for all hospital server communications.

• HL7 FHIR R4 standard for EHR interoperability.

• All data at rest encrypted with AES-256; data in transit secured via TLS 1.3.

• Role-Based Access Control (RBAC) governing who can view patient data or modify system configurations.

AI Model Development

Generative AI Methods: Three AI Components working together in the AIPMA Pipeline

Component A: GAN for Synthetic ICU Data Generation

A TimeGAN (Time-series GAN) architecture is used to generate realistic synthetic vital sign sequences representing rare but clinically important deterioration patterns such as sepsis progression, respiratory failure, and cardiac arrhythmia onset. The GAN is trained on de-identified ICU datasets (e.g., MIMIC-IV). Synthetic data is then used to balance the training dataset for the anomaly detection model, preventing the model from being biased toward normal states.

Component B: Autoencoder for Real-Time Anomaly Detection (EDGE)

A compact LSTM-based autoencoder is trained to reconstruct normal vital sign windows. During inference, it processes 60-second multivariate windows (heart rate, SpO2, respiratory rate, blood pressure, temperature) and computes a reconstruction error score. Windows that exceed a dynamically calibrated threshold, accounting for patient-specific baselines, are flagged for deeper analysis. This model is quantized to INT8 using TensorFlow Lite to meet the processing constraints of edge hardware and achieve sub-100ms inference latency.

Component C: Fine-Tuned LLM for Explainable Alert Generation

A parameter-efficient fine-tuned version of a clinical LLM (e.g., BioMedLM or a quantized LLaMA variant fine-tuned on medical literature and ICU notes) serves as the reasoning engine. When an anomaly is confirmed, the LLM receives a structured prompt containing: current vital sign values, the direction and rate of change of each parameter, relevant patient context from the EHR (age, weight, admitting diagnosis, active medications, allergies), and the anomaly score. The LLM generates a plain-language alert in under two seconds, citing which vital signs are most concerning and recommending a priority level (Watch / Escalate / Emergency).

Autonomous Decision-Making Logic

The Agent operates in a 3 -Tier Escalation Framework.

Alert Level	Trigger Condition	Autonomous Action	Human Role
WATCH	Mild anomaly score; single parameter deviation	Log event; update dashboard trend indicator	Clinician reviews at next check-in
ESCALATE	Moderate anomaly; multi-parameter correlation	Push notification to nurse; generate LLM summary	Nurse responds within 5 minutes
EMERGENCY	High anomaly score; critical threshold crossed	Audible alarm + push alert to physician on call; auto-document in EHR	Immediate physician response

Crucially, the agent does not autonomously administer medications or modify treatment. All autonomous actions are informational and advisory. Final clinical decisions always rest with human caregivers, consistent with FDA guidelines for Software as a Medical Device (SaMD).

Tools Needed for Implementation

• **TensorFlow Lite:** for edge deployment of the quantized autoencoder model.

• **Edge Impulse:** for model optimization, profiling, and deployment pipeline to edge hardware.

• **Hugging Face Transformers:** for fine-tuning and serving the LLM component.

• **PyTorch:** for TimeGAN training and prototyping.

• **MIMIC-IV database:** as the primary de-identified ICU dataset for training and evaluation.

• **Node-RED:** for orchestrating the data flow pipeline between edge nodes and the hospital server.

Security & Ethical Considerations

Data Security & Privacy

• All patient data is processed on-premise at the edge or hospital server. No identifiable patient information is transmitted to external cloud services without explicit consent and anonymization.

• HIPAA compliance is enforced through technical safeguards: end-to-end encryption (AES-256 at rest, TLS 1.3 in transit), access logs, and automatic session timeouts.

• Federated Learning is proposed for any future model retraining, models are updated using gradients derived from local data, never sharing raw patient records between institutions.

• Device authentication via X.509 certificates ensures only registered edge nodes can publish to the MQTT broker.

• Intrusion Detection System (IDS) rules are configured on the hospital's network to detect anomalous device behavior, such as unexpected data volume spikes or unauthorized access attempts.

Ethical Considerations & Bias Mitigation

• **Training data bias:** MIMIC-IV is predominantly drawn from a single academic medical center with specific patient demographics. The model must be validated across diverse patient populations before clinical deployment. Bias audits should be conducted quarterly.

• **Alarm fatigue:** Excessive false positives erode clinician trust in AI alerts. The system is designed to minimize false positive rates through multi-parameter correlation and patient-specific baselines rather than fixed thresholds.

• **Explainability requirement:** Every AI-generated alert must include a plain-language rationale. Black-box outputs are not acceptable in clinical settings. The LLM component is specifically responsible for meeting this requirement.

• **Equitable care:** Alert sensitivity will be tested across patient subgroups (age, sex, race, comorbidity burden) to ensure the system does not systematically underperform for any group.

• **Human oversight:** The system is designed as a decision support tool, not an autonomous treatment agent. No autonomous action changes patient care without human approval.

Implementation & Testing Plan

Phased Deployment Plan

If this system were to be implemented, the following phased approach would be used:

• **Phase 1 – Data Pipeline Setup (Weeks 1–3):**

Deploy MQTT broker and edge nodes in the simulated ward environment. Configure HL7 FHIR interfaces with mock EHR data. Validate sensor-to-edge data ingestion at full load (20 simultaneous patients).

• **Phase 2 – Model Training & Optimization (Weeks 4–7):**

Train TimeGAN on MIMIC-IV data. Augment dataset and train LSTM autoencoder. Quantize autoencoder to TensorFlow Lite INT8. Fine-tune LLM on ICU note subset. Validate inference latency targets (<100ms edge, <2s LLM).

• **Phase 3 – Integration & Pipeline Testing (Weeks 8–10):**

Integrate all three AI components into the Node-RED pipeline. Test end-to-end alert generation. Integrate with mock EHR via REST API. Conduct A/B alert comparison against threshold-only system.

• **Phase 4 – Security Hardening & Ethical Audit (Weeks 11–12):**

Penetration testing of MQTT broker and REST endpoints. HIPAA compliance audit. Bias evaluation across patient subgroups.

• **Phase 5 – Pilot Simulation & Performance Review (Weeks 13–15):**

Simulate 72-hour ICU operation with synthetic patient streams. Measure sensitivity, specificity, false positive rate, and alert latency. Document findings for clinical validation submission.

Test Scenarios

Test Category	Scenario Description	Success Metric
Functional – Normal	20 patients with stable vitals over 4 hours	Zero false-positive ESCALATE/EMERGENCY alerts
Functional – Deterioration	Inject synthetic sepsis trajectory into 3 patient streams	Alert generated within 60 seconds of deterioration onset
Stress Test	40 concurrent patient streams at 5x normal data rate	Edge inference latency stays below 100ms; no data loss
Security – Auth	Attempt to connect unauthorized device to MQTT broker	Connection refused; intrusion event logged within 5 seconds
Security – Injection	Send malformed or out-of-range vital sign values	Input validation rejects packet; system continues normally
Edge Offline Resilience	Simulate hospital server going offline for 10 minutes	Edge node continues local alerting autonomously; data queued for sync on reconnect
Bias Evaluation	Run synthetic patient cohorts across age, sex, and racial demographic groups	Alert sensitivity variance < 5% across subgroups

Performance Metrics

- **Alert Sensitivity (Recall):**

Proportion of true deterioration events that generate an alert. Target: >92%.

- **Alert Specificity:**

Proportion of normal periods that do not generate false alerts. Target: >85%.

- **Alert Latency:**

Time from deterioration onset (defined by clinical ground truth) to alert delivery. Target: <90 seconds.

- **Edge Inference Latency:**

Time for autoencoder to process one 60-second window. Target: <100ms on target hardware.

- **LLM Summary Generation Time:**

Time to produce a clinician-ready alert narrative. Target: <2 seconds.

- **System Uptime:**

Target 99.9% availability (including edge node failover).

Challenges, Trade-Offs & Lessons Learned

Key Challenges Identified

• **Data scarcity for rare events:** ICU deterioration events (e.g., cardiac arrest) are inherently rare, creating severe class imbalance. This was addressed conceptually through GAN-based synthetic data generation, but GAN training instability remains a known challenge that would require careful hyperparameter tuning and validation of synthetic sample quality.

• **Edge hardware constraints:** Running an LSTM autoencoder plus an LLM on edge hardware is computationally demanding. The architecture resolves this by placing the computationally heavy LLM inference on the hospital server rather than the edge node, preserving edge resources for the latency-critical autoencoder.

• **EHR integration complexity:** Hospital EHR systems (Epic, Cerner, etc.) use proprietary APIs alongside HL7 FHIR. Full integration would require vendor-specific development and IT security approvals, representing a significant real-world barrier.

• **Clinical trust and validation:** AI systems in clinical settings face rigorous validation requirements (FDA SaMD classification, IRB approval for trials). Achieving clinical acceptance requires extensive prospective validation beyond what simulations can provide.

Design Trade-Offs

• **On-premise edge vs. cloud inference:** Choosing edge-only processing for PHI ensures HIPAA compliance and eliminates latency from network round-trips, but limits computational resources. This trade-off was deliberately accepted to prioritize privacy and reliability.

• **Explainability vs. model performance:** More complex deep learning models (e.g., transformer-based detectors) might achieve higher sensitivity but are harder to explain. The chosen autoencoder + LLM hybrid balances performance with explainability.

• **Alert sensitivity vs. specificity:** Tuning for higher sensitivity catches more real events but increases false positives (alarm fatigue). The dynamic threshold approach attempts to balance this using patient-specific baselines rather than a fixed system-wide cutoff.

Lessons Learned in the Planning Process

• Designing for healthcare AI requires understanding clinical workflows, not just technical performance metrics. An alert that arrives too late, is too vague, or is too frequent is clinically useless even if the underlying model is accurate.

• Security and privacy are not features to add at the end, they must be architected into every layer from the beginning, especially given HIPAA obligations.

• Generative AI (GAN + LLM) adds substantial value not just as a modeling technique but as a bridge between raw sensor data and human-understandable clinical decision support. The explainability function of the LLM component is arguably as important as its predictive role.

• Autonomous does not mean unsupervised. Designing the AIPMA as a decision support tool rather than an autonomous treatment agent was a deliberate ethical and regulatory choice that should guide all future healthcare AI deployments.

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